Data Engineering ASSIGNMENT 03

P. Vaishnavi
24NIUIA4431
CSD / 2nd year.

1. Explain the concept & importance of "Change Data Capture" (CDC) in the context of Data integration.

Change Data Capture (CDC) is a technique used to identify and capture changes (insert, update, delete) in a source database and apply those changes to another system such as a data warehouse or analytics platform.

Ex:- Student Table :- → StuId | Name | Branch | Address | Phno

Fee Table → StuId | Name | Category | Fee

Academics Table → StuId | Name | CGPA

⇒ If a student's CGPA changes from 7.5 to 8.2, CDC captures only that update instead of copying the entire academics table to another system.

StuId	Name	Branch	CGPA
101	Ravi	CSE	7.5
102	Kavi	ECE	8.0

Suppose Ravi improves his C.G.P.A 7.5 to 8.0

Updated table :-

StuId	Name	Branch	CGPA
101	Ravi	CSE	8.2
102	Kavi	ECE	8.0

Without CDC, the system copies the entire table again to another system.

With CDC, only the change event is captured.

Operation	StuId	Column Changed	Old Value	New Value
UPDATE	101	CGPA	7.5	8.2

Importance of CDC in Data Integration :-

* Efficiency: Only changed data is transferred, reducing processing and storage cost.

* Real-time updates:- Systems like dashboards and analytics platforms get updated quickly.

* Scalability :- Works well even when databases contain millions of records

* Data consistency :- Keeps multiple systems synchronized with the source database.

2. Briefly discuss about the data base logs, insert only and different types of databases - non-relational, key-value stores, document-stores, wide column and graph databases.

Database logs

(Transaction logs) record every change made in a database such as INSERT, UPDATE and DELETE operations. They help in data recovery, auditing and replication.

Logs keep a history of database activities.

Operation	Table	Record ID	Change
INSERT	Student	101	Added Ravi
UPDATE	Student	101	CGPA 7.5 → 8.2

Insert-Only Databases :-

Data is never updated or deleted. Instead, new records are added for every change. This keeps a complete history of data changes.

Non-Relational Databases (NoSQL)

Non-relational databases store data in flexible formats rather than strict tables. They are designed for large-scale, distributed systems.

Major types include:-

Key-value Stores:-

Data is stored as a simple key and value pair, similar to dictionary.

Key	Value
user101	Ravi
user102	Asha

Document Stores

Data is stored as documents (JSON format) with flexible structure.

Example :-

```
{
"id" : 101,
"name" : "Ravi",
"branch" : "CSE".
}
```

Wide column Databases

A wide column database is a type of NoSQL database where data is stored in columns instead of fixed rows. Columns are grouped into column families, and each row can have different columns.

Row Key	Name	Branch	CGPA
101	Ravi	CSE	8.2
102	Asha	ECE	–

Graph Databases:

graph database is a type of NoSQL database that stores data using nodes (entities) and edges (relationships). It focuses on how different pieces of data are connected.

Ex:-

Node 1	Relationship	Node 2
User A	follows	User B
User B	friends with	User C

3. a) Differentiate between Single and multitier, monolithic and microservices architectures.

b) How do Operational and Technical Architecture work together to create a "good" data architecture.

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Single-Tier Architecture	Multitier Architecture
Everything (data, application, UI) is on a single server	System is divided into multiple layers like presentation, business & data.
High risk; if one part fails the entire system fails	Low risk; failure in one layer does not affect the whole system.
Limited performance; heavy tasks can slow the application	Better performance because tasks are distributed across layers.
Used for local testing or small prototype systems	Used for real-world production application

Monolithic Architecture	Microservices Architecture
Single large application where all components are tightly integrated.	Application is divided into small independent services
Deployed as a single unit	Services are deployed independently
hard to scale specific parts of the system	Easy to scale individual services.
Failure may affect the entire system	Failure is isolated to a specific service.

b) Operational architecture focuses on how data systems operate and run daily, including data pipelines, monitoring, backup and recovery processes.

Technical Architecture focuses on the technologies and tools used, such as databases, storage systems, processing frameworks and integration platforms.

* Both work together to create a good data architecture by ensuring that the system is scalable, reliable, secure and easy to maintain.

Operational architecture ensures smooth functioning and recovery during failures, while technical architecture provides the appropriate technologies and system structure to support data processing efficiently.